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Every figure in this document is recomputed from `results/task2b/*.txt` at build time; none is typed by hand.

1. Headline

Baseline LRU, 25M warmup + 25M simulation, GCC 7.5.0, three traces. Speedup is the geomean of per-trace IPC ratios; MPKI is the arithmetic mean over the three equal-length traces.

policy	geomean speedup	mean L2C MPKI	mean instr MPKI
LRU (baseline)	1.0000	16.3356	10.1050
plain RRIP	1.0004	15.8080	9.8467
static I-RRIP (Task 2a)	1.0138	14.7590	6.2219
dynamic I-RRIP (Task 2b)	1.0068	14.5693	7.2543

Two facts carry the result:

1. **Dynamic I-RRIP has the lowest mean L2C MPKI of all four policies (14.5693).** It even beats static I-RRIP (14.7590), because it avoids static I-RRIP's one regression. On trace1, static I-RRIP raises L2C MPKI +8.7% over LRU by protecting instructions a workload that does not reuse them; the dynamic policy sees that in its leaders, stays in RRIP mode, and lands at +0.5%.
2. **It does not beat static I-RRIP on IPC (1.0068 vs 1.0138),** because each trace is internally homogeneous – a run-time switch has no phase change to exploit, so the best it can do is converge to the right static choice while still paying the leader-set tax.

2. The duel chose correctly on all three traces

trace	prioritising (I-RRIP) share	switches	PSEL (mid 512)	correct?
trace1	12.1%	20	519	yes - I-RRIP hurts trace1, stays RRIP
trace2	82.9%	39	501	yes - I-RRIP helps trace2, adopts it
trace3	68.0%	14	512	yes - marginal, PSEL ends ~neutral

Switch counts of 14-39 over tens of millions of accesses are the duel settling around a stable answer, not thrashing.

3. The PDF-mandated leader-set tracking

The policy tracks instruction vs non-instruction accesses and misses to the RRIP leader sets (reported cumulatively, not from the decayed windowed counters the selector uses):

trace	instr acc	instr miss	data acc	data miss
trace1	4971	3487	5575	3376
trace2	10634	5910	8647	4550
trace3	2421	1824	3314	2254

4. Storage

The dynamic policy stores 2 RRPV bits/line (forced: 1 bit collapses the two modes into one) plus a small selector. Total **2058 bytes** – about double static I-RRIP (1024 B), still a third smaller than the LRU it replaces (3072 B). The whole increase is the second RRPV bit; the duel itself is negligible.

5. Which policy is best?

- *Hand-picking one policy per trace*: static I-RRIP wins the two it suits.
- *Committing one policy to an unknown mix*: the dynamic policy wins – best mean L2C MPKI and no catastrophic regression (worst +0.96%, versus static I-RRIP's +8.7% on trace1). That robustness is what set-duelling exists to buy, and what the Task 2b design goal rewards.

The dynamic policy does exactly what Task 2b asks – reads behaviour and chooses correctly on all three traces. It does not top the IPC chart because these traces have no intra-trace phases to exploit, a property shown by the low switch counts above, not a defect in the policy. Beating the standalones on IPC is the Task 3 bar (SPP + dynamic I-RRIP), which extends this duel.